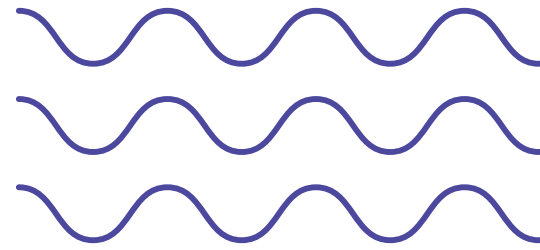




**Fusion Software
Institute**

DATA SCIENCE



Unleash the power of data science at Fusion Software Institute. Dive into the world of analytics, machine learning, and AI to unlock insights and drive innovation.



www.fusion-institute.com



1. INTRODUCTION

This Data Science course Syllabus, in collaboration with Top MNC, features exclusive hackathons, masterclasses, and Ask-me- anything sessions for the best training experience. This Data Science certification Course Syllabus provides hands-on exposure to key technologies including R, Python, Machine Learning, Tableau, PowerBI, Hadoop, and Spark via live interaction with practitioners, practical labs, and Live Industry Projects. Create a dynamic career in the field of Data Science and Machine Learning with Fusion Technology Master's program Offered by Certified Experts. Our curriculum covers all the concepts of Data Science enabling you to become an expert in these two complementary disciplines that organizations are seeking globally.

2. TOOLS

- **Anaconda**
- **Jupyter notebook**
- **Pycharm**
- **VM ware**
- **AWS Account Creation**
- **ETL (Extract Transform Load)**



PYTHON SYLLABUS

INTRODUCTION OF PYTHON

- What is Python?
- Why Python?
- Installing Python Python IDEs
- Jupyter Notebook Overview

HANDS-ON-EXERCISE

- Installing Python idle for windows, Linux and
- Creating “**Hello World**” code.

PYTHON BASICS

- Introduction
- Variables
- Python Basic Data types
- Lists
- String Slicing, Indexing
- Operator
- IF statements Loops
- Dictionaries
- Tuples
- Functions I) Functions arguments
II) Lambda Function III) Built in function
- Modules, Packages anaconda
- Array

HANDS-ON-EXERCISE- CONSTRUCTING OPERATORS

- Practice and Quickly learn Python necessary skills by solving simple questions and problems.

- Practice and Quickly learn Python necessary skills by solving simple questions and problems.
- How Python uses indentation to structure a program, and how to avoid some common indentation errors.
- You executed to make simple numerical lists, as well as a few operations you can perform on numerical lists, tuples, dictionary and set.

PYTHON PACKAGES

- Pandas Numpy
- Sci-kit Learn
- **Mat-plot library Hands-on-Exercise:** Installing jupyter notebook for windows, Linux and Installing numpy, pandas and matplotlib

IMPORTING DATA

- Reading CSV files
- Saving in Python data
- Loading Python data objects
- Writing data to CSV file

HANDS-ON-EXERCISE:

- To generate data sets and create visualizations of that data. You learned to create simple plots with matplotlib, and you saw how to use a scatter plot to explore random
- You learned to create a histogram with Pygal and how to use a histogram to explore the results of rolling dice of different
- Generating your own data sets with code is an interesting and powerful way to model and explore a wide variety of real-world
- As you continue to work through the data visualization projects that follow, keep an eye out for situations you might be able to model with

SQL SYLLABUS

INTRODUCTION OF DATABASE

- List the features of MySQL Database.
- Discuss the basic design, theoretical, and physical aspects of a relational database.
- Categorize the different types of SQL statements.
- Describe the data set used by the course.
- Log on to the database using SQL Developer environment.
- Save queries to files and use script files in SQL Developer

HANDS-ON-EXERCISE:

- List the features of MySQL Database.
- Work with MySQL database tools.
- Understand and work with language features.

RETRIEVE DATA USING THE SQL SELECT STATEMENT :

- List the capabilities of SQL SELECT statements.
- Generate a report of data from the output of a basic SELECT statement.
- Select All Columns
- Select Specific Columns
- Use Column Heading Defaults
- Use Arithmetic Operators
- Understand Operator Precedence.
- Learn the DESCRIBE command to display the table structure.

HANDS-ON-EXERCISE:

- Individual statements in SQL scripts are commonly terminated by a line break (or carriage return) and a forward slash on the next line, instead of a semicolon.
- You can create a SELECT statement, terminate it with a line break, include a forward slash to execute the statement, and save it in a script file.

LEARN TO RESTRICT AND SORT DATA:

- Write queries that contain a **WHERE** clause to limit the output retrieved
- List the comparison operators and logical operators that are used in a **WHERE** clause.
- Describe the rules of precedence for comparison and logical operators.
- Use character string literals in the **WHERE** clause.
- Write queries that contain an **ORDER BY** clause to sort the output of a **SELECT** statement.
- Sort output in descending and ascending order.

HANDS-ON-EXERCISE:

- Creating the queries in a compound query must return the same number of columns.
- Create corresponding columns in each query must be of compatible data types.
- **ORDER BY**; it is, however, permissible to place a single **ORDER BY** clause at the end of the compound query.

USAGE OF SINGLE-ROW FUNCTIONS TO CUSTOMIZE OUTPUT

- Describe the differences between single-row and multiple-row functions.
- Manipulate strings with character functions in the **SELECT** and **WHERE** clauses.
- Manipulate numbers with the **ROUND**, **TRUNC**, and **MOD** functions
- Perform arithmetic with date data.
- Manipulate dates with the **DATE** functions.

HANDS-ON-EXERCISE:

- Create the distinction is made between single-row functions, which execute once for each.
- row in a dataset, and multiple-row functions, which execute once for all the rows in a data- set.



INVOKE CONVERSION FUNCTIONS AND CONDITIONAL EXPRESSIONS.

- Describe implicit and explicit data type conversion.
- Use the **TO_CHAR**, **TO_NUMBER**, and **TO_DATE** conversion functions.
- Nest multiple functions.
- Apply the **NVL**, **NULLIF**, and **COALESCE** functions to data.
- Use conditional **IF THEN ELSE** logic in a **SELECT**.

HANDS-ON-EXERCISE:

- We create and discuss the **NVL** function, which provides a mechanism to convert null values into more arithmetic-friendly data values.



AGGREGATE DATA USING THE GROUP FUNCTIONS

- Use the aggregation functions in **SELECT** statements to produce meaningful reports
- Divide the data into groups by using the **GROUP BY** clause
- Exclude groups of data by using the **HAVING** clause

HANDS-ON-EXERCISE:

- Group functions operate on aggregated data and return a single result per group.
- These groups usually consist of zero or more rows of data.



DISPLAY DATA FROM MULTIPLE TABLES USING JOINS

- Write **SELECT** statements to access data from more than one table
- View data that generally does not meet a join condition by using outer joins
- Join a table by using a self-join

USE SUBQUERIES TO SOLVE QUERIES

- Describe the types of problems that subqueries can solve
- Define sub-queries
- List the types of sub-queries

HANDS-ON-EXERCISE:

- Write a query that uses subqueries in the column projection list.
- Write single-row and multiple-row subqueries



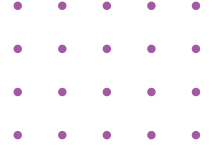
DATA MANIPULATION

STATEMENTS:

- Describe each **DML** statement
- Insert rows into a table
- Change rows in a table by the **UPDATE** statement
- Delete rows from a table with the **DELETE** statement
- Save and discard changes with the **COMMIT** and **ROLLBACK** statements.
- Explain read consistency.

HANDS-ON-EXERCISE:

Expressions and create expose a vista of data manipulation possibilities through the interaction of arithmetic and character operators with column or literal data, or a combination of the two.



USE OF DDL STATEMENTS TO CREATE AND MANAGE TABLES

- Categorize the main database objects
- Review the table structure.
- List the data types available for columns
- Create a simple table.
- Decipher how constraints can be created at table creation.

MANIPULATING DATA

- Selecting rows/Observations Rounding Number
- Selecting columns/fields Merging data
- Data aggregation



HANDS-ON-EXERCISE- CONSTRUCTING OPERATORS

- As you gain experience with CSV and JSON files, you'll be able to process almost any data you want to analyze.
- Most online data sets can be downloaded in either or both of these. From working with these formats, you'll be able to learn other data formats as well.





MACHINE LEARNING SYLLABUS

MACHINE LEARNING SUPERVISED LEARNING

LINEAR REGRESSION:

- Linear Equation
- Slope o Intercept

LOGISTIC REGRESSION:

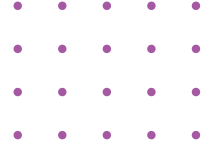
- ODDS ratio Probability of success
- Probability of failure Bias Variance Tradeoff ROC cur
- Bias Variance Tradeoff

RIDGE LASSO:

- we've reviewed the main ways to approach the problem of modeling data using simple and definite



UNSUPERVISED LEARNING



- K-Means
- Hierarchical Clustering

SVM

- Support Vectors
- Hyperplanes
- 2-D Case
- Linear Hyperplane

SVM KERNAL

- Linear
- Radial
- Polynomial

OTHER MACHINE LEARNING ALGORITHMS

- K- Nearest Neighbour
- Naïve Bayes Classifier
- Decision Tree - CART
- Decision Tree - C50
- Random Forest
- Adaboosts
- XGBoosts



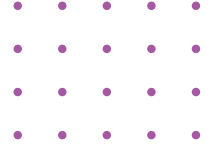
HANDS-ON-EXERCISE:

- We have covered the simplest but still very practical machine learning models in an eminently practical way to get us started on the complexity.
- Where we will cover several regression techniques, it will be time to go and solve a new type of problem that we have not worked on, even if it's possible to solve the problem with clustering methods (regression), using new mathematical tools for approximating unknown values.
- In it, we will model past data using mathematical functions, and try to model new output based on those modelling

STATISTICS IN MACHINE LEARNING

CENTRAL TENDENCY

1. Mean o Median
2. Mode
3. Skewness
4. Normal Distribution



PROBABILITY BASICS:

- What does it mean by probability?
- Types of Probability

STANDARD DEVIATION

- Data deviation & distribution
- Variance

BIAS VARIANCE TRADE-OFF

- Underfitting
- Overfitting

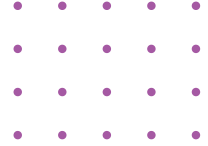
DISTANCE METRICS

- Euclidean
- Distance

OUTLIER ANALYSIS

- What
- is an Outlier?
- Inter





STATISTICS IN MACHINE LEARNING

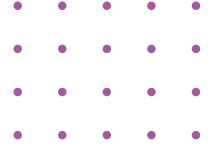
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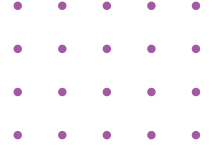
BIAS VARIANCE TRADE-OFF

- Underfitting
- Overfitting

DISTANCE METRICS

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- Distance





OUTLIER ANALYSIS

- What
- is an Outlier?
- Inter
- Quartile
- Range Box &
- whisker plot Upper
- Whisker Lower
- Whisker Scatter
- plot

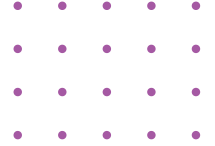
MISSING VALUE TREATMENT

- What is NA?
- Central Imputation
- KNN imputation
- Domification

CORRELATION:

- Pearson correlation
- positive & Negative correlation





HANDS-ON-EXERCISE:

- Compute probability in a situation where there are equally-likely outcomes.
- Apply concepts to cards and dice.☒
- Compute the probability of two independent events both occurring.
- Compute the probability of either of two independent events occurring.
- Do problems that involve conditional probabilities
- Calculate the probability of two independent events occurring.
- List all permutations and combinations
- List all permutations and combinations



ERROR METRICS IN ML

CLASSIFICATION:

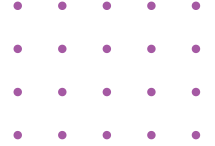
- Confusion
- Matrix
- Precision
- Recall
- Specificity
- F1
- Score

REGRESSION:

- MSE
- RMSE
- MA E

HANDS-ON-EXERCISE:

- State why the z' transformation is necessary Compute the standard error of z
- Compute a confidence interval on p The computation of a confidence interval
- Estimate the population proportion from sample proportions
- Apply the correction for continuity



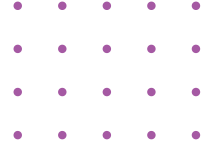
DEEP LEARNING SYLLABUS

MODULE-15: DEEP LEARNING ALGORITHMS

- ANN – Artificial Neural Network
- RNN – Recurrent Neural Network
- CNN – Convolutional Neural Network
- Activation functions
- Loss and cost functions
- Keras, Tensorflow framework
- Optimization function

HANDS-ON-EXERCISE:

- We took a very important step towards solving complex problems together by means of implementing our first neural .
- Now, the following architectures will have familiar elements, and we will be able to extrapolate the knowledge acquired on this chapter, to novel



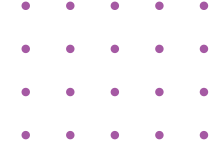
ARTIFICIAL INTELLIGENCE AND NATURAL LANGUAGE PROGRAMMING

NLP

- Text Pre-processing
- Noise Removal
- Lexicon Normalization
- Lemmatization
- Stemming

FUSION SOFTWARE
INSTITUTE





TEXT TO FEATURES (FEATURE ENGINEERING)

- Syntactical Parsing
- Dependency Grammar Part of Speech Tagging Entity Parsing
- Named Entity Recognition Topic Modelling
- N-Grams TF – IDF

ARTIFICIAL INTELLIGENCE

- Complete ML, DL, NLP
- Image processing
- Feature extraction
- Object detection and Recognition
- Convolutional neural networks(CNNs)
- Transfer Learning
- NEURAL NETWORK

HANDS-ON-EXERCISE:

- Provided, you will even be able to create newly customized
- As our models won't be enough to solve very complex problems, in the following chapter, our scope will expand even more, adding the important dimension of time to the set of elements included in our generalization.

WHY CHOOSE US?

- 100% Job Placement Assistance
- Industry Experts as a Trainer
- Resume Preparation
- Weekly Mock Interviews
- Digital Classroom
- 2 Real-time Projects
- Expert Faculty
- Expert Instructors
- Hands-On Learning
- Career Support
- Innovative Curriculum
- State-of-the-Art Facilities
- Career Path Support
- Industry-Relevant Courses
- Community Connection
- Exclusive Networking Opportunities
- Career Services Beyond Graduation
- Industry-Recognized Certifications
- Provide Internship Program
- We Provide Soft Skill Corporate Training



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